

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack} (\text{frame } e_0 l) F \dots) FS_I \dots))$

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack} (\text{frame } E[(\text{async}/\text{lambda} (x \dots_1) e) v \dots_1] l) F \dots) FS_I \dots)) \longrightarrow (t_1 \sigma_1 Q (FS_0 \dots (\text{stack} (\text{frame } e_{x_{\text{async}}}) (\text{frame } E[(\text{lambda} () e)] l) F \dots) FS_I \dots))$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack} (\text{frame } v l) F \dots) FS_I \dots))$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack} (\text{frame } (\text{throw } v_{\text{err}}) l) F \dots) FS_I \dots))$

$(t_0 \sigma_0 (FS_0 \dots (\text{stack} (\text{frame } E[(\text{run} (\text{lambda} () e))] l) F \dots) FS_I \dots))$

$(t_0 \sigma Q (FS_0 \dots (\text{stack} (\text{frame } E[(\text{await} (\text{task } x_{\text{async}}))] l) F \dots) FS_I \dots))$

$(t_0 \sigma Q (FS_0 \dots (\text{stack} (\text{frame } E[(\text{await} (\text{task } x_{\text{async}}))] l) F \dots) FS_I \dots))$

$(t_0 \sigma_0 Q (FS_0 \dots (\text{stack current-frame } F \dots) FS_I \dots))$

where $(\text{async? } l)$, $v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{async}}]$, $(\text{pending } \dots) = \text{task-state}[v_{\text{obj}}]$, $\text{current-frame} = (F \dots)$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack} (\text{frame } E[(\text{os}/\text{io natural } v)] l) F \dots) FS_I \dots))$

$(t_0 \sigma_0 Q_0 (FS_0 \dots (\text{stack} (\text{frame } E[(\text{os}/\text{resolve} (\text{task } x_{\text{async}}) t_{\text{resolve}} v)] l) F \dots) FS_I \dots))$

$(t_0 \sigma Q (FS_0 \dots (\text{stack} (\text{frame } E[(\text{os}/\text{resolve } v_{\text{task}} t_{\text{resolve}} v)] l) F \dots) FS_I \dots))$

$(t_0 \sigma Q_0 (FS_{\text{main}} FS_0 \dots (\text{stack}) FS_I \dots))$

$(t_0 \sigma Q ((\text{stack} (\text{frame } E[(\text{block} (\text{task } x_{\text{async}}))] l)) FS_{\text{rest}} \dots))$

$\longrightarrow (t_1 \sigma_1 Q (FS_0 \dots (\text{stack} (\text{frame } e_l l) F \dots) FS_I \dots))$

where $(\text{not value?}[e])$, $(\sigma_l e_l) = \Downarrow \text{base}[e]$

$\longrightarrow (t_1 \sigma_1 Q_l (FS_0 \dots (\text{stack } F \dots) FS_I \dots))$

where $(\text{async? } l)$, $x_{\text{async}} = l$

$\longrightarrow (t_1 \sigma_1 Q_l (FS_0 \dots (\text{stack } F \dots) FS_I \dots))$

where $(\text{async? } l)$, $x_{\text{async}} = l$, $v_{\text{obj}} = \text{lookup}[\sigma, x_{\text{async}}]$

$\longrightarrow (t_1 \sigma_1 Q_l (FS_0 \dots (\text{stack} (\text{frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_I \dots))$

$\longrightarrow (t_1 \sigma Q (FS_0 \dots (\text{stack} (\text{frame } E[v] l) F \dots) FS_I \dots))$

where $(\text{done } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]]$

$\longrightarrow (t_1 \sigma Q (FS_0 \dots (\text{stack} (\text{frame } E[(\text{throw } v)] l) F \dots) FS_I \dots))$

where $(\text{failed } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]]$

$\longrightarrow (t_1 \sigma_1 Q (FS_0 \dots (\text{stack } F \dots) FS_I \dots))$

$\longrightarrow (t_1 \sigma_1 Q_l (FS_0 \dots (\text{stack} (\text{frame } E[(\text{task } x_{\text{async}})] l) F \dots) FS_I \dots))$

$\longrightarrow (t_1 \sigma_1 Q_l (FS_0 \dots (\text{stack } F \dots) FS_I \dots))$

where $(\geq t_0 t_{\text{resolve}})$, $v_{\text{obj}} = \text{lookup}[\sigma_0, x_{\text{async}}]$

$\longrightarrow (t_1 \sigma Q (FS_0 \dots (\text{stack} (\text{frame } E[(\text{os}/\text{resolve } v_{\text{task}} t_{\text{resolve}} v)] l) F \dots) FS_I \dots))$

where $(< t_0 t_{\text{resolve}})$

$\longrightarrow (t_1 \sigma Q_l (FS_{\text{main}} FS_0 \dots (\text{stack } F_{\text{head}}) FS_I \dots))$

where $\text{all-busy?}[FS_0, \dots]$, $(F_{\text{head}} Q_l) = \text{q-pop}[FS_0, \dots]$

$\longrightarrow (t_1 \sigma Q ((\text{stack} (\text{frame } E[v] l)) FS_{\text{rest}} \dots))$

where $(\text{sync? } l)$, $(\text{done } v) = \text{task-state}[\text{lookup}[\sigma, x_{\text{async}}]]$